Assignment 3

Data Processing Pipeline:
CSV Cleaning, Feature Engineering & Pytest

Contents

1. Code: pipeline.py (load, clean, add_features)
2. Code: test_pipeline.py (one pytest test for clean)
3. Screenshot: running the pipeline
4. Screenshot: pytest result (1 passed)
5. Output file: students_clean.csv

07 November 2026

Code: pipeline.py (part 1 of 2)

pipeline.py

```
"""Simple data processing pipeline: load -> clean -> add_features -> save."""

from pathlib import Path

import pandas as pd

HERE = Path(__file__).parent
INPUT_FILE = HERE / "students_messy.csv"
OUTPUT_FILE = HERE / "students_clean.csv"

# Ages written as words in the raw data
WORD_TO_NUMBER = {
    "eighteen": 18,
    "nineteen": 19,
    "twenty": 20,
    "twenty one": 21,
    "twenty two": 22,
    "twenty three": 23,
    "twenty four": 24,
    "twenty five": 25,
}


def load(path=INPUT_FILE):
    """Load the raw student data from a CSV file."""
    return pd.read_csv(path)


def clean(df):
    """Return a cleaned copy of the student data."""
    df = df.copy()

    # 1. Trim extra spaces from text columns
    for col in ["name", "city"]:
        df[col] = df[col].str.strip()

    # 2. Same spelling for every city (karachi / KARACHI -> Karachi)
    df["city"] = df["city"].str.title()

    # 3. Turn ages written as words into numbers ("twenty" -> 20)
    df["age"] = df["age"].astype(str).str.strip().str.lower().replace(WORD_TO_NUMBER)
    df["age"] = pd.to_numeric(df["age"], errors="coerce")

    # 4. Drop rows that have no name
    df = df.dropna(subset=["name"])
    df = df[df["name"] != ""]

    # 5. Remove duplicate rows
    df = df.drop_duplicates()

    # 6. Fill missing scores with the median score
    df["score"] = df["score"].fillna(df["score"].median())

    df["age"] = df["age"].astype(int)
    return df.reset_index(drop=True)


def grade(score):
    """Turn a score into a letter grade."""
    if score >= 85:
        return "A"
    if score >= 70:
        return "B"
    if score >= 50:
        return "C"
    return "F"

```

Code: pipeline.py (part 2 of 2)

pipeline.py

```
    print(f"Loaded {len(raw)} rows from {Path(input_path).name}")
    print(raw.to_string(), "\n")

    cleaned = clean(raw)
    print(f"After clean(): {len(cleaned)} rows")

    final = add_features(cleaned)
    final.to_csv(output_path, index=False)
    print(final.to_string(), "\n")
    print(f"Saved {len(final)} rows to {Path(output_path).name}")
    return final


if __name__ == "__main__":
    run_pipeline()
```

Code: test_pipeline.py

test_pipeline.py

```
import pandas as pd

from pipeline import clean


def test_clean():
    messy = pd.DataFrame(
        {
            "name": ["Ali", "Omar", "Omar", None, "Sara"],
            "city": ["Karachi", "KARACHI ", "KARACHI ", "Lahore", " lahore"],
            "age": ["21", "twenty", "twenty", "23", "22"],
            "score": [80.0, None, None, 66.0, 60.0],
        }
    )

    result = clean(messy)

    # Duplicate Omar row and the row with no name are removed
    assert result["name"].tolist() == ["Ali", "Omar", "Sara"]
    # City names are trimmed and use the same capitals
    assert result["city"].tolist() == ["Karachi", "Karachi", "Lahore"]
    # Word ages become numbers
    assert result["age"].tolist() == [21, 20, 22]
    # Missing score is filled with the median (of 80 and 60)
    assert result["score"].isna().sum() == 0
    assert result.loc[1, "score"] == 70.0
```

Screenshot: running the pipeline

The pipeline loads students_messy.csv, cleans it, adds features, and saves students_clean.csv.

Windows PowerShell

```
PS D:\testing\practise-demo\assignment3> py pipeline.py
Loaded 7 rows from students_messy.csv
    name      city     age  score
0    Ali   Karachi      21   80.0
1   Sara   karachi      22   92.0
2   Omar  KARACHI   twenty    NaN
3   Omar  KARACHI   twenty    NaN
4   Hina    Lahore      20   72.0
5    NaN    Lahore      23   66.0
6  Bilal    lahore      22   38.0

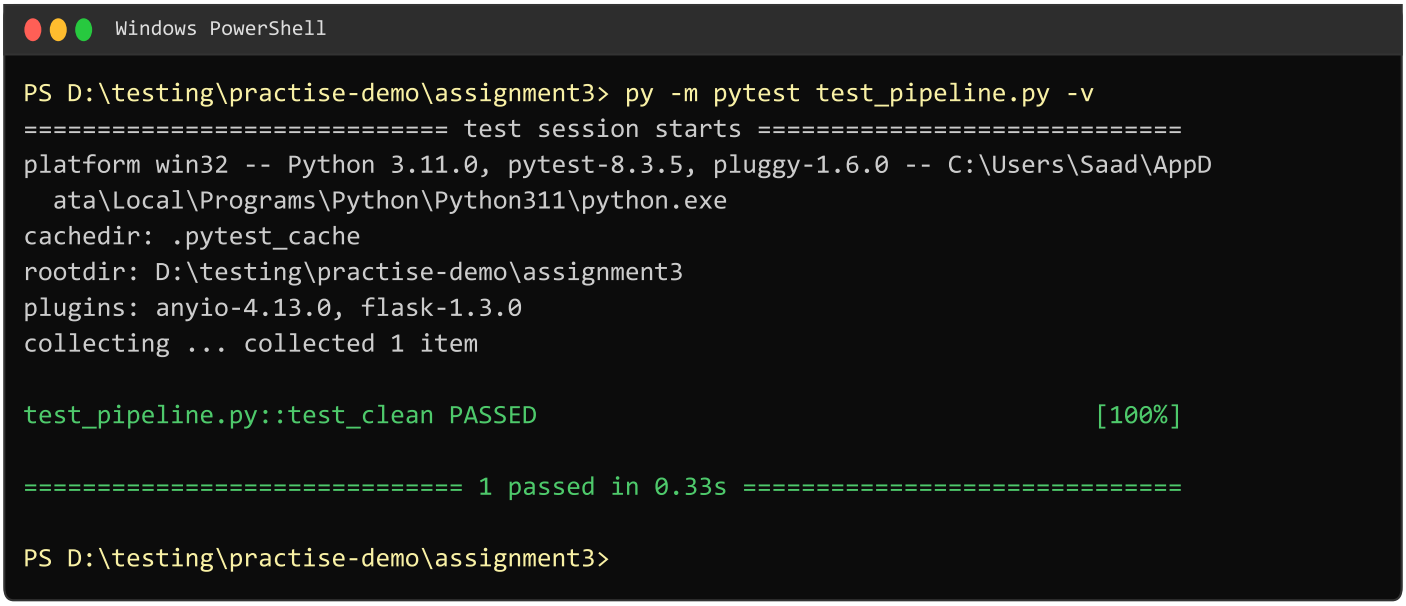
After clean(): 5 rows
    name     city  age  score  passed grade
0    Ali  Karachi   21   80.0    True     B
1   Sara  Karachi   22   92.0    True     A
2   Omar  Karachi   20   76.0    True     B
3   Hina   Lahore   20   72.0    True     B
4  Bilal   Lahore   22   38.0   False     F

Saved 5 rows to students_clean.csv

PS D:\testing\practise-demo\assignment3>
```

Screenshot: pytest result

The test for clean() passes: 1 passed.



Output file: students_clean.csv

5 rows after removing the duplicate and the row with no name. New columns: passed, grade.

students_clean.csv

```
name,city,age,score,passed,grade
Ali,Karachi,21,80.0,True,B
Sara,Karachi,22,92.0,True,A
Omar,Karachi,20,76.0,True,B
Hina,Lahore,20,72.0,True,B
Bilal,Lahore,22,38.0,False,F
```

Cleaning steps in clean():

- Trimmed extra spaces from name and city
- City names in the same case (karachi / KARACHI -> Karachi)
- Age written as a word converted to a number (twenty -> 20)
- Dropped the row with no name
- Removed the duplicate Omar row
- Filled Omar's missing score with the median score (76.0)

Features added in add_features():

- passed: True if score >= 50
- grade: A (>= 85), B (>= 70), C (>= 50), F (below 50)